

Reference Reading List

Prioritized by thesis chapter — for the TSUKIMI-coupled thermal pipeline
Ramon Gregorio — April 2026

★★★ = Read this week (critical path)

★★ = Read within 2 weeks

★ = Read when relevant

Chapter 2: Thermal Model Physics

2.2 – Continuous H-parameter model

★★★ Hayne, P.O. et al. (2017) Global regolith thermophysical properties of the Moon from the Diviner Lunar Radiometer Experiment.

J. Geophys. Res.: Planets, 122, 2371–2400. doi:10.1002/2017JE005387

THE foundational paper. Read Sections 2–4 for the H-parameter equations. Your implementation starts here. Also read the Supplementary for the specific heat polynomial.

★★★ Hayne, P.O. (2017) heat1d: Thermal model for planetary science applications.

GitHub: github.com/phayne/heat1d. Also available on arXiv: 1711.00977

The reference code. Compare your Crank-Nicolson against his Fourier-matrix solver. Pay attention to the specific heat coefficients in the code vs. the paper.

★★ Vasavada, A.R. et al. (2012) Lunar equatorial surface temperatures and regolith properties from the Diviner Lunar Radiometer Experiment.

J. Geophys. Res., 117, E00H18. doi:10.1029/2011JE003987

The predecessor two-layer model. Useful for understanding the historical progression.

2.3 – Low-temperature conductivity correction

★★★ Martinez, A. & Siegler, M.A. (2021) A global thermal conductivity model for lunar regolith at low temperatures.

J. Geophys. Res.: Planets, 126, e2021JE006829. Code: zenodo.org/records/12586656

Your second most important paper. Read Sections 2–3 for the low-T K_c modification. Download their MATLAB code from Zenodo and port the key functions to Python.

★★ Woods-Robinson, R., Siegler, M.A. & Paige, D.A. (2019) A model for the thermophysical properties of lunar regolith at low temperatures.

J. Geophys. Res.: Planets, 124, 1989–2011. doi:10.1029/2019JE005955

Shows that standard specific heat fits fail below 40 K. Important for PSR modeling.

2.4 – Microphysical model (comparison)

★★ Bürger, J. et al. (2024) A microphysical thermal model for the lunar regolith: investigating the latitudinal dependence of regolith properties.

J. Geophys. Res.: Planets, 129, e2023JE008152. doi:10.1029/2023JE008152

The newest approach — replaces H-parameter with grain-size physics. Read to understand why constant properties fail at all latitudes. Consider as a third switchable option if time permits.

★ Gundlach, B. & Blum, J. (2012) Outgassing of icy bodies in the solar system – II: heat transport in dry, porous surface dust layers.

Icarus, 219, 618–629. doi:10.1016/j.icarus.2012.03.013

2.7 – Validation data (Apollo, Chang'E-4, ChaSTE)

★★ **Nagihara, S. et al. (2018)** Examination of the long-term subsurface warming observed at the Apollo 15 and 17 sites utilizing the newly restored heat flow experiment data from 1975 to 1977.

J. Geophys. Res.: Planets, 123, 1125–1139. doi:10.1029/2018JE005579

Restored Apollo HFE data. Confirms astronaut-induced 2–4 K warming. Your sanity-check validation target.

★★ **Xiao, Z. et al. (2022)** Thermophysical properties of the regolith on the lunar far side revealed by the in situ temperature probing of the Chang'E-4 mission.

National Science Review, 9, nwac175. doi:10.1093/nsr/nwac175

In situ data at Von Kármán crater. Surface density 471 kg/m³ — much lower than Apollo. Good cross-validation point.

★★ **Durga Prasad, K. et al. (2025)** High-resolution thermal profiles of lunar regolith over a southern high latitude location using in situ observations from ChaSTE/Chandrayaan-3 lander.

MNRAS, 538, 2330. Also: Sci. Rep. (2025), thermal conductivity paper.

First high-latitude (69.4°S) in situ K measurement. $K = 0.0115\text{--}0.0124$ W/m/K at 80 mm.

2.7 – Lab simulant measurements (supporting)

★ **Sakatani, N. et al. (2018)** Thermal conductivity of lunar regolith simulant JSC-1A under vacuum.

Icarus, 309, 13–24. doi:10.1016/j.icarus.2018.02.027

Laboratory K values for JSC-1A: $(2\text{--}3)\times 10^{-3}$ W/m/K at 240–340 K.

★ **Wache, S. et al. (2025)** Investigation of the thermal conductivity of dry and icy lunar regolith simulants.

J. Geophys. Res.: Planets. doi:10.1029/2025JE009084

Icy regolith analogs. Validates Gundlach-Blum model with ice addition.

Chapter 3: Illumination and Shadow Modeling

3.1–3.2 – DEM and horizon-based illumination

★★★ **Mazarico, E. et al. (2011)** Illumination conditions of the lunar polar regions using LOLA topography.

Icarus, 211, 1066–1081. doi:10.1016/j.icarus.2010.10.030

THE reference for horizon-based illumination. Your ray-tracer follows this method. Validate your illumination maps against their published products.

★★ **Barker, M.K. et al. (2021)** Improved LOLA elevation maps for south pole landing sites: error estimates and their impact on illumination conditions.

Planet. Space Sci., 203, 105119. doi:10.1016/j.pss.2020.105119

5 m DEM products and illumination uncertainty quantification. Use their PGDA data products for your 20 m pipeline.

★★ **Gläser, P. et al. (2018)** Illumination conditions at the lunar poles: implications for future exploration.

Planet. Space Sci., 162, 170–178.

20 m illumination products. Good comparison target.

★ **Hao, W. et al. (2023)** A high-precision horizon-based illumination modeling method for the lunar surface using pyramidal LOLA data.

Icarus, 394, 115436. doi:10.1016/j.icarus.2022.115436

Operates on LOLA point clouds rather than gridded DEMs. Advanced approach.

3.3 – Secondary illumination and view factors

★★★ **Schorghofer, N. (2024)** Planetary-Code-Collection: Thermal, ice evolution, and exosphere models for planetary surfaces.

GitHub: github.com/nschorgh/Planetary-Code-Collection

Your primary reference implementation for Topo3D view factors. Read the Topo3D/ directory README and fieldofview.f90 source files.*

★★ **Mahanti, P. et al. (2022)** View factor based computation of secondary illumination within lunar permanently shadowed regions.

IEEE Geosci. Remote Sens. Lett., 19, 8027004.

Formal view-factor approach validated by ShadowCam. 60 m resolution.

★★ **Potter, S.F. et al. (2023)** Fast hierarchical low-rank view factor matrices for thermal irradiance on planetary surfaces.

J. Comput. Phys.: X, 17, 100130. arXiv: 2209.07632

HODLR compression for scalable radiosity. Read if you need to go beyond nearest-neighbor.

★ **Kloos, J.L. et al. (2021)** Illumination conditions within permanently shadowed regions at the lunar poles: implications for in-situ passive remote sensing.

Acta Astronautica, 180, 25–34.

Shows ~0.03% of PSR elements receive no scattered light. Quantifies secondary illumination.

3.4 – Shadow dynamics and PSR evolution

★★ **Williams, J.-P. et al. (2019)** Seasonal polar temperatures on the Moon.

J. Geophys. Res.: Planets, 124, 2505–2521. doi:10.1029/2019JE006028

Decade of Diviner data. Seasonally cold-trapped areas 2.8–4.3× larger than PSRs. Primary polar temperature validation target.

★ **O'Brien, P. & Byrne, S. (2022)** Double shadows: mapping the extent of doubly permanently shadowed regions.

Planet. Sci. J., 3, 258.

Coldest places on the Moon. Only 5.37 km² at south pole.

★ **Schörghofer, N. & Rufu, R. (2023)** Past extent of lunar permanently shadowed areas.

Science Advances, 9, eadh4302.

PSRs are geologically young (~1.8 Gyr average). Constrains ice accumulation time.

Chapter 4: Spatial Thermal Mapping Pipeline

★★ **Paige, D.A. et al. (2010)** Diviner Lunar Radiometer observations of cold traps in the Moon's south polar region.

Science, 330, 479–482. doi:10.1126/science.1187726

Original Diviner polar thermal maps. Your benchmark for surface temperature products.

★★ **Powell, T.M. et al. (2023)** High-resolution nighttime temperature and rock abundance mapping of the Moon using the Diviner Lunar Radiometer Experiment with a model for topographic removal.

J. Geophys. Res.: Planets, 128, e2022JE007532.

13-year Diviner reprocessing. 3.5× resolution improvement. Use RDR v3.1 data.

★★ **King, O. et al. (2020)** The Oxford 3D thermophysical model with application to PROSPECT/Luna 27 study landing sites.

Planet. Space Sci., 182, 104790.

O3DTM combines heat1d with 3D scattering. Reduces errors from ~10 K to ~2 K. Your quality target.

★★ **Hayes, A.G. et al. (2024)** Topography-enhanced ultra-cold trapping at the LCROSS impact site.

J. Geophys. Res.: Planets, 129, e2023JE007925.

1 m/pixel thermal modeling. Highest resolution published. Shows micro cold traps at 20–30 K.

★ **Williams, J.-P. et al. (2017)** The global surface temperatures of the Moon as measured by the Diviner Lunar Radiometer Experiment.

Icarus, 283, 300–325.

Global Diviner temperature compilation. Reference for equatorial/mid-lat validation.

★ **Ambily, S. et al. (2026)** Diurnal temperature variability at latitudinally distinct sites on the Moon.

Earth and Space Science, doi:10.1029/2025EA004568.

Very recent. Multi-latitude diurnal analysis.

Chapter 5: TSUKIMI RTM Coupling and Jacobian Inversion

5.1 – THz/microwave RTM forward modeling

★★★ Wang, Z. et al. (2024) Terahertz emission modeling of lunar regolith.

Remote Sensing, 16(21), 4037. doi:10.3390/rs16214037

TSUKIMI team's own paper. Matrix doubling RTM for THz with volume scattering. This is the forward model your thermal profiles feed into.

★★★ Feng, J., Siegler, M.A. & Hayne, P.O. (2020) New constraints on thermal and dielectric properties of lunar regolith from LRO Diviner and CE-2 microwave radiometer.

J. Geophys. Res.: Planets, 125, e2019JE006130.

Gold-standard thermal→RTM coupling pipeline. Reconciles Diviner IR with CE-2 microwave. Your template for the coupling architecture.

★★ Yang, Y. et al. (2022) A calibrated lunar microwave radiative transfer model based on satellite observations.

Remote Sensing, 14(21), 5501.

23-layer temperature profiles from Hayne's model coupled to RTM at 23–183 GHz. Shows how to format thermal profiles for RTM input.

★★ Wei, G. et al. (2019) A new method for simulation of lunar microwave brightness temperatures and evaluation of Chang'E-2 MRM data using thermal constraints from Diviner.

J. Geophys. Res.: Planets, 124, doi:10.1029/2018JE005858.

Diviner-constrained thermal model as RTM upper boundary. Same philosophy as your pipeline.

5.2–5.4 – Jacobian and inversion methodology

★★ Gong, X. et al. (2014) Inversion of dielectric properties of the lunar regolith media with temperature profiles using Chang'E microwave radiometer observations.

IEEE Geosci. Remote Sens. Lett., 12, 384–388.

Regolith property inversion from microwave T_B using thermal model constraints. Closest published analogue to your Jacobian approach.

★★ Feng, J. et al. (2021) Reconciling the infrared and microwave observations of the lunar south pole: a study on subsurface temperature and regolith density.

J. Geophys. Res.: Planets, 126, e2021JE006623.

Joint Diviner–MRM inversion at south pole. Shows density/conductivity ambiguity that the Jacobian analysis can resolve.

★★ Zhong, Z. et al. (2023) Retrieval of lunar polar heat flow from Chang'E-2 microwave radiometer and Diviner observations.

Frontiers in Astronomy and Space Sciences, 10, 1179558.

Heat flow inversion combining thermal + microwave models. South polar heat flow ~4.9 mW/m².

★ Weng, F. & Liu, Q. (2003) Satellite data assimilation in numerical weather prediction models. Part I: forward radiative transfer and Jacobian models in cloudy atmospheres.

J. Atmos. Sci., 60, 2633–2646.

General Jacobian formulation for RTM-coupled inversion. Atmospheric context but the mathematics (tangent-linear, adjoint) transfers directly to your regolith problem.

5.5 – THz penetration depth and scattering

★ Chen, L. et al. (2024) Experimental analysis of terahertz wave scattering characteristics of simulated lunar regolith surface.

Remote Sensing, 16(20), 3819.

Lab measurements of THz scattering from regolith simulant. Constrains loss tangent at THz.

★ TSUKIMI mission website Technology overview.

<https://sites.google.com/tsukimi.one/english/technology>

Mission parameters, frequency bands, expected penetration depths.

Chapter 6: Ice Survivability Mapping

★★★ Schorghofer, N. & Williams, J.-P. (2020) Mapping of ice storage processes on the Moon with time-dependent temperatures.

Planet. Sci. J., 1, 54. doi:10.3847/PSJ/ab9a4c

THE ice stability methodology paper. Time-averaged sublimation rates vs peak-T thresholds. 24% larger cold-trap area. Directly follow their algorithm.

★★★ Hayne, P.O., Aharonson, O. & Schörghofer, N. (2021) Micro cold traps on the Moon.

Nature Astronomy, 5, 169–175. doi:10.1038/s41550-020-1198-9

Micro cold traps at 1 cm–1 km scale. Total cold-trapping area revised to ~40,000 km². Your 20 m resolution can resolve these.

★★ Davidsson, B.J.R. & Hosseini, S. (2025) Determining the conditions for thermal stability of global near-surface water ice on the Moon.

MNRAS, 538, 809–819.

Buried ice stable to ~131 K surface T at 1 m depth. Desiccated surface layer analysis.

★★ Schorghofer, N. & Williams, J.-P. (2024) Supervolatile cold traps and water ice stability on the Moon.

Icarus, 416, 116086.

Multi-volatile stability: CO₂ <60 K, NH₃ <81 K. Comprehensive maps.

★★ Schörghofer, N. (2025) Current theories of lunar ice.

arXiv: 2502.06056. Review article.

Excellent overview. Covers delivery, retention, loss, and observational evidence. Read as a crash course before writing Chapter 6.

★ Li, S. et al. (2026) ShadowCam finds scarce surface ice in PSRs.

Science Advances (2026).

Negative result: no widespread surface ice at >20–30 wt%. Underscores need for subsurface thermal models — motivates your work.

★ Landis, M.E. et al. (2022) Spatial distribution and thermal diversity of surface volatile cold traps at the lunar poles.

Planet. Sci. J., 3, 39.

Cold-trap classification by thermal environment. Useful for comparing your maps.

Open-Source Tools and Datasets

Codes to download and study

★★★ **heat1d** — github.com/phayne/heat1d

Hayne's reference 1D thermal solver. Python/C/MATLAB. MIT license. Compare your solver output against this.

★★★ **Planetary-Code-Collection** — github.com/nnschorgh/Planetary-Code-Collection

Schorghofer's Fortran/Python toolkit. Includes Topo3D (view factors, horizon computation), Crank-Nicolson solver, and ice stability models. Port the Topo3D view-factor code.

★★ **Martinez-Siegler low-T model** — zenodo.org/records/12586656

MATLAB code for the low-temperature conductivity correction. Port to Python.

★ **O3DTM** — github.com/tw7044/O3DTM

Oxford 3D thermophysical model. MATLAB. Combines heat1d with terrain scattering.

★ **KRC** — github.com/NASA-Planetary-Science/krc

Kieffer's Fortran model. 50+ year heritage. Primarily Mars but applicable to Moon.

Datasets to download

★★★ **LOLA 20 m South Pole DEM** — pgda.gsfc.nasa.gov/products/90

LDEM_80S_20MPP_ADJ.TIF (2.6 GB). Track-adjusted, co-registered. Also 10 m and 5 m options available.

★★★ **Diviner Polar Cumulative Products** — pds-geosciences.wustl.edu (search Diviner GDR L3)

Williams et al. (2019) polar temperature maps. 240 m bins, 0.25-hr local time resolution. Your primary validation dataset.

★★ **Diviner bolometric temperature maps** — diviner.ucla.edu/data

Global bolometric T at 0.5°/0.25 hr. Also rock abundance and thermal inertia maps.

★★ **Apollo HFE restored data** — PDS4 format via pds-geosciences.wustl.edu

Nagihara et al. (2018) restored 1975–1977 data. Your sanity-check target.

★ **SLDEM2015** — pgda.gsfc.nasa.gov/products/54

60 m LOLA-Kaguya merged DEM. Only $\pm 60^\circ$ latitude — not for polar work but useful for equatorial validation.

Suggested Reading Order (First 3 Weeks)

Week 1 (Apr 14–20): Hayne et al. (2017) Sections 2–4 + heat1d code. Martinez & Siegler (2021) Sections 2–3 + Zenodo code. Wang et al. (2024) for TSUKIMI RTM context.

Week 2 (Apr 21–27): Schorghofer & Williams (2020) for ice stability methodology. Feng et al. (2020) for thermal→RTM coupling pipeline. Mazarico et al. (2011) for illumination algorithm. Download and explore Planetary-Code-Collection Topo3D.

Week 3 (Apr 28–May 4): Bürger et al. (2024) for microphysical comparison understanding. Gong et al. (2014) and Feng et al. (2021) for Jacobian/inversion concepts. Hayne et al. (2021) micro cold traps. Williams et al. (2019) for Diviner polar validation products.

Ongoing: Schorghofer (2025) arXiv review as a reference for ice theory. Hayes et al. (2024) for high-resolution modeling benchmark. Powell et al. (2023) for Diviner data reprocessing context.